
Question #: 1

An investigator studying nutrition asks participants to keep a food and exercise diary for a month. An entry from a 43-year-old man weighing 73 kg (160 lb) shows that he consumed 300 g of carbohydrates, 80 g of proteins, 90 g of fats, 20 g of ethanol, and 25 g of dietary fiber on the previous day. His daily energy expenditure was approximately 2400 Calories. Which of the following would be the best conclusion from these data?

A. He has to decrease the dietary fiber content for optimal health
B. To maintain the present weight, he needs to consume an additional 400 Calories
C. He is consuming 200 Calories more than his requirement
D. He has to reduce the intake of dietary protein to maintain nitrogen balance
✓E. Dietary fats are providing about 33% of his total calorie intake

Rationale: The caloric content of the diet is 1200 (Calories from carbohydrates) + 320 (Calories from proteins) + 810 (Calories from Fats) + 140 (Calories from alcohol) = 2470 Calories. Percentage of Calories from fats = 810/2470 = 32.7% = 33% of Calories from fats. He is consuming 70 Calories in excess of his requirement.

Question #: 2

An investigator studying nutrition prepares meals containing amino acids of various compositions as the sole contribution of dietary protein. These meals are fed to a group of normal adult female volunteers who are 25-30 years old. A protein deficient in which of the following amino acids would most likely result in a negative nitrogen balance in these research subjects?

A. Cysteine
✓B. Leucine
C. Tyrosine
D. Asparagine
E. Glutamate

Rationale: To maintain nitrogen equilibrium in adults it is important to consume proteins which contain all the essential amino acids. Consumption of proteins deficient in one of the essential amino acids results in a negative nitrogen balance. The essential amino acids are PVT TIMHALL (Phenylalanine, valine, tryptophan, threonine, isoleucine, methionine, histidine, arginine, leucine and lysine). Histidine and arginine are essential during periods of growth for example,during childhood and adolescence, and are usually not required for adults.

Question #: 3

A 25-year-old man comes to the physician for a follow-up examination. He says he gained 10-kg (22-lbs) during the past 3 years. He now weighs 90 kg (198 lb), his waist-to-hip ratio is 1.4; and BMI is 34 kg/m^2. Which of the following metabolic changes most likely occurred following this patient's change in weight?

A. Decrease in the size of adipocytes
✓B. Increase in the secretion of leptin
C. Activation of the hunger center
D. Increase in the secretion of ghrelin
E. Decrease in the number of adipocytes

Rationale: Leptin is a hormone released by adipocytes. An increase in body weight results in an increased secretion of leptin. Leptin inhibits the hunger center. There is increased ghrelin secretion when a person is fasting. Ghrelin is secreted from the stomach.

Question #: 4

An investigator discovers an aberrant karyotype in an unaffected mother of a child who has an intellectual disability and dysmorphology. The karyotype of the mother shows a fusion of the long arms of two non-homologous acrocentric chromosomes and she has 45 total chromosomes. Which of the following best explains the appearance of this karyotype?

A. Reciprocal translocation
✓B. Robertsonian translocation
C. Meiosis I nondisjunction
D. Chromosome inversion
E. Autosomal triploidy
F. Imprinting error

Rationale: Robertsonian translocation describes how two acrocentric chromosomes may fuse to form one chromosome. A person who carries this translocation might appear unaffected, but their children will be at risk for chromosomal imbalance.

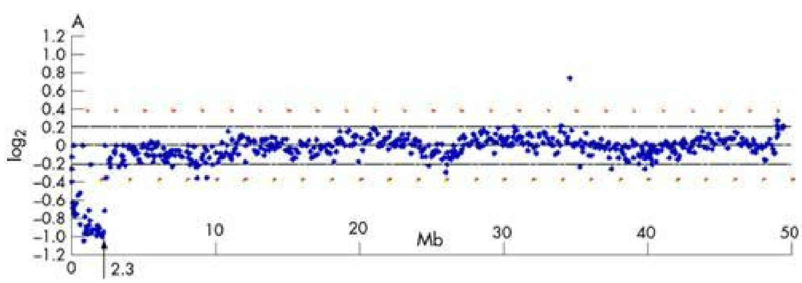
Question #: 5

A 11-year-old boy is brought to the physician because of development delay, intellectual disability, and facial anomalies. The child's G-banded metaphase karyotype appeared normal, and showed no detectable chromosomal abnormalities including, inversions, insertions, deletions, or duplications. A different genetic showed an alteration on chromosome 5, as shown in the figure. The G-band karyotype did not detect the abnormality because it

✓A. Offers less resolution than array comparative genome hybridization
B. Does not offer information regarding gene expression
C. May only be used with chromosomes isolated from mitotic cells
D. Is not able to detect deletions on a chromosome
E. Relies on polymerase chain reaction technology
F. Requires the hybridization of a radioactively labeled probe

Rationale: This question is asking you to compare two genomic diagnostic technologies: the standard karyotype, and the microarray CGH. Array CGH has higher resolution for copy number variation compared to karyotype, so array CGH is able to detect, and provide a diagnosis, more frequently than karyotype.

Attachment:



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Question #: 6

A 13-year-old boy is brought to the physician by his adoptive parents for a follow-up examination because of a lifetime history of mild learning disabilities, mild behavior problems, and acne. Standard karyotype of 40 metaphase cells shows that the boy has a mosaic karyotype consisting of 46,XY, (30 cells) and 47,XY+16 (10) cells. Which of the following is most likely responsible for this patient's mixed karyotype?

A. Maternal nondisjunction at meiosis I
B. Maternal nondisjunction at meiosis II
✓C. Mitotic nondisjunction after fertilization
D. Paternal nondisjunction at meiosis II
E. Paternal nondisjunction at meiosis I

Rationale: In general a mosaic karyotype can be caused by two mechanisms. It is thought that the more common mechanism is trisomy rescue after fertilization. In trisomy rescue, a gamete is produced that has two copies of a chromosome (24 chromosomes total, instead of 23). When this is fertilized by the other, normal gamete, there will be 47 total chromosome. If a post-fertilization event occurs where a cell loses (sheds) one of these extra chromosomes, we say "trisomy rescue" has occurred. Note that if this occurs with chromosome 15, there is a possibility of creating uniparental disomy. If it is paternal uniparental disomy 15 (meaning lack of maternal contribution), Angelman syndrome occurs. If maternal uniparental disomy occurs (meaning lack of paternal contribution), then Prader-Willi syndrome is likely to be caused.

The other (second) general mechanism that may cause a mosaic karyotype would be a nondisjunction event that occurs during a mitotic cell division. By "mitotic" cell division, we mean that the nondisjunction occurred following a fertilization by a normal (or typical) sperm and a normal (or typical) ovum. In this case, two cell types would combine to form the eventual embryo and fetus: one with a normal karyotype, and one that is abnormal.

Question #: 7

A 32-year-old woman comes to the physician for follow up after giving birth to a child with dysmorphic facial features that included a depressed nasal bridge and upturned palpebral fissures. The karyotype of the child and polymorphic marker analysis of the family are shown. Based on these data, which of the following most likely caused the aneuploidy condition in the child?

A. Robertsonian translocation involving chromosome 21
B. Meiosis I nondisjunction in the mother
C. Meiosis I nondisjunction in the father
✓D. Meiosis II nondisjunction in the mother
E. Meiosis II nondisjunction in the father

Rationale: The polymorphic marker analysis shows that the baby has one allel marker from the father, and one allele marker from the mother, but the AMOUNT of DNA that is shown from the child in the lower band looks DOUBLE all the other bands suggesting that there was nondisjunction of sister chromatids during meiosis !! of oogenesis.

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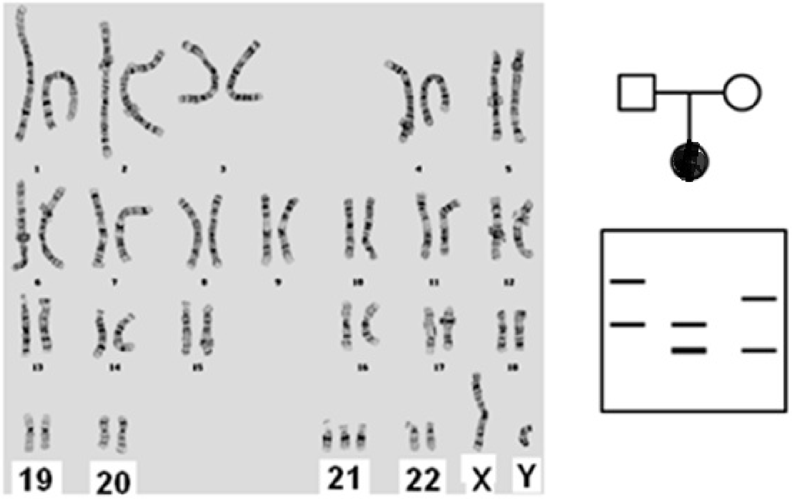

ExamSoft ID 21843 post mt online Q karyotype trisomy 21 and markers.jpg

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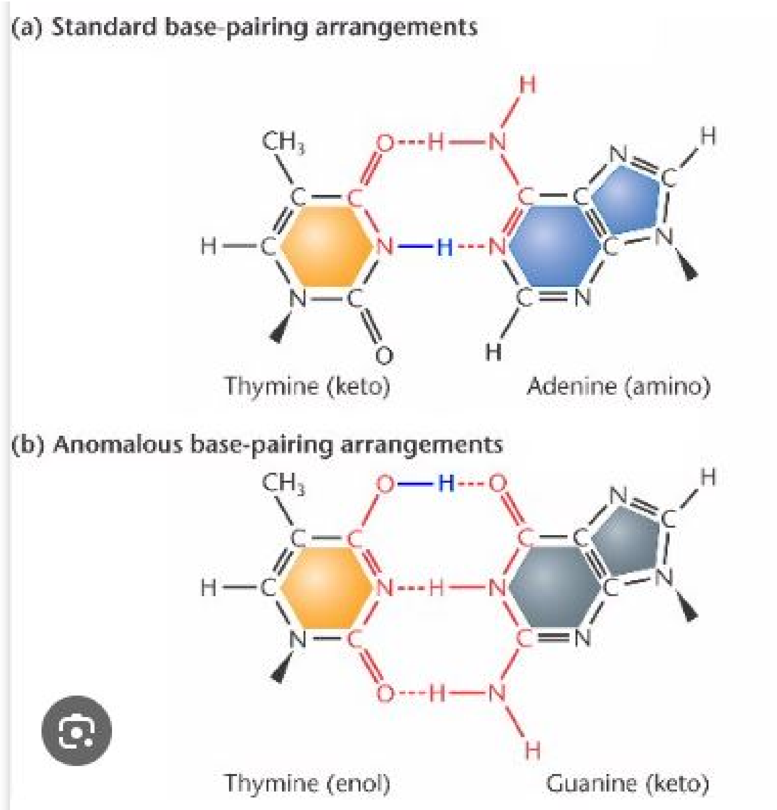
Question #: 8

A scientist studying genome stability creates an assay to detect DNA repair by the proof-reading mechanism. The assay measures repair of mis-incorporated bases that were caused by tautomeric shifts of thymine as shown in the attached figure. In which of the following stage of the cell cycle did this repair most likely occur?

A. G1 phase
✓B. S phase
C. G2 phase
D. M phase
E. G0 phase

Rationale: A temporary change in base structure by isomerization leads to tautomer. Tautomers are bases which can exist in the keto and enol forms. Thymine typically exists in the keto form and will base-pair with adenine during DNA synthesis, but a transient shift of thymine to the enol form will allow for base-pairing with guanine. The mis-match can only occur during DNA synthesis when the thymine is free to undergo a spontaneous rearrangement of the nitrogen bases to form a tautomeric shift.

Attachment:



Tautomer.JPG

Question #: 9

A 16-year-old girl is brought to the physician by her mother because of a severe sunburn. Her sunburn is due to 3 hours of exposure to ultraviolet light, a component of the electromagnetic spectrum of sunshine. Because of this exposure, DNA damage accumulates in her skin cells. Which of the following repair systems is most likely responsible for the correction of this damaged DNA before replication?

A. Mismatch repair
B. Base excision repair
C. Non-homologous end joining
✓D. Nucleotide excision repair
E. Proofreading

Rationale: Sun sensitivity is a clinical feature of Xeroderma pigmentosa. Briefly, UV exposure can lead to pyrimidine dimer formation. This damage can be repaired in human cells by the activity of 9 enzymes in a pathway called nucleotide excision repair. If a person has mutation in any of these 9 different genes in the pathway, it may lead to this genetic disorder.

Question #: 10

A researcher studying cancer is investigating the signal transduction pathway that cells use to regulate cell division. A diagram of the major cell proliferative pathway shown. Which of the following must occur to cause a normally functioning Ras protein to become active in the context of this pathway?

A. Ras mus be phosphorylated on a tyrosine
B. Ras must disengage from the membrane
C. Ras must be bound to GDP
D. Receptor must become dephosphorylated
✓E. Growth factor hormone must bind to receptor

Rationale: Upon binding to hormone, the growth factor receptor (shown here as a receptor tyrosine kinase) becomes activated causing Ras to release GDP from its guanosine nucleotide binding sit. GTP is now free to enter this site, causing Ras to become active. Ras has an intrinsic timer function, so it hydrolyzes the GTP at some point forming GDP, and turning off downstream signaling. Ras is not phosphorylated on tyrosine amino acid side chains. Ras stays attached to the inner leaflet of the plasma membrane as a peripherally bound protein. Ras must be bound to GTP to be activated. Typically, Ras becomes activated when the receptor is activated (and phosphorylated) in response to hormone binding.

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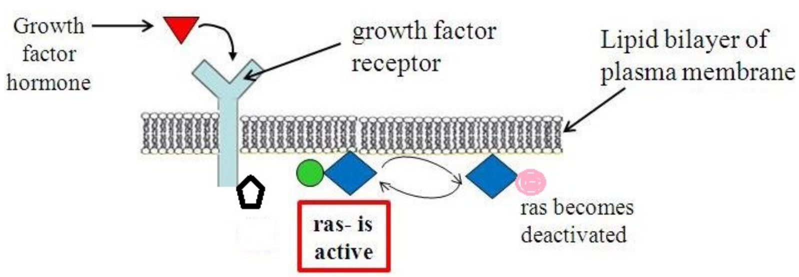

RAS cartoon_a.jpg

Question #: 11

A 9-month-old boy is brought to the opthamologist by his parents after observing strange white reflections in both eyes in all photographs taken with a flash in the last month. Physical examinition with an opthalmoscope shows a 2-milimeter white fluffy retinal mass in both eyes. Which of the following cellular events is most likely asociated with the formation of the tumors in this patient?

A. Activation of p53
B. Loss of 3' to 5' proof reading of DNA polymerase
C. Activation of the G2/M checkpoint
D. Release of cytochrome c from the mitochondria
✓E. Promotion of the G1/S transition

Rationale: Retinoblastoma is a rare type of eye cancer, develops in early childhood and one of 3 children develop cancer in both eyes. The first sign is a visible whiteness in the pupil called "cat's eye reflex" or leukocoria as shown in the figure. Standard treatment may include chemotherapy and radiation therapy followed by surgery.

Attachment:

Rb.JPG

Question #: 12

A 33-year-old scientist studying DNA repair subjects an in-vitro culture of human epithelial cells to x-ray radiation. The ionizing radiation induces DNA strand breakage. Which of the following events would most likely occur in these cells as a result of this damage?

A. The BCL-2 gene will be expressed at high levels
B. The perforin/granzyme apoptotic pathway will be activated
C. The p53 protein will be inactive and unphosphorylated
✓D. The amount of BAX protein will be elevated in the cell
E. The survival pathway will be activated

Rationale: X-ray radiation induces DNA strand breakage. When DNA damage occurs, cellular mechanisms are activated to either repair the damage or initiate cell death is the damage can not be repaired. DNA damage can activate the stability of p53 which will then induce the expression of BAX, which will translocate to the mitochondria, then promote the formation of pores in the outer mitochondrial membrane, cytochrome c and other pro-apoptotic factors are released into the cytosol and finally this triggers the activation of caspases which lead to programmed cell death (apoptosis).

Question #: 13

A 12-year-old girl is brought to the physician by her parents for follow-up because of the progressive appearance of neurofibromas. Physical examination shows she now has more than 200 distinct formations of these neoplasms, and Lisch nodules on the iris of yer eyes. Her father and paternal grandmother had a similar condition. Which of the following is the most probable normal function of the gene that is mutated in this patient?

A. DNA repair
B. Proto-oncogene
✓C. Tumor suppressor
D. Apoptosis regulator
E. Cell cycle checkpoint regulator

Rationale: Neurofibromatosis type 1 (NF1) is an autosomal dominant disorder that is an extremely variable multisystem disease. The progression and severity may differ in affected family members with the same pathogenic variant. The NF1 gene coded for a protein called neurofibromin that helps regulate cell growth and is considered a classical example of a tumor suppressor gene. Tumor suppressors normally prevent cells from growing and dividing too rapidly

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Question #: 14

A researcher studying colon cancer finds that 2 –5% of colon cancer cases are heritable. One familial type is called HNPCC (hereditary non-polyposis colon cancer). Compilation of the genomic structure and mutation profile of cells obtained from 100 resected HNPCC tumors shows a specific pathognomonic signature. Which of the following best describes the function of the pathway that, when mutationally inactivated, leads to the development of this type of cancer?

A. Activation of G2/M checkpoint

B. Signaling through APC

✓C. DNA mismatch repair

D. Nucleotide excision repair

E. Initiation of apoptosis

Rationale: Lynch syndrome, or HNPCC is a type of hereditary colon cancer. HNPCC is characterized by a single, relatively fast progressing tumor. Biopsy of cells from an HNPCC tumor will show microsatellite instability due to accumulation of mutation in areas of the genome that have simple sequence repeats (SSR). The number of repeats in the SSR regions tend to change rapidly in people who have loss of function of the DNA mismatch repair pathway.

Question #: 15

A researcher studying cancer in an in vitro model system, uses site directed mutagenesis to alter one allele of a gene in a cell. The missense base-pair change leads to an amino acid substitution in the encoded protein. In response to the mutation, the cell line is found to attain the properties of oncogenic transformation and unregulated proliferation. Which of the following genes is most likely to have been mutated by the researcher?

A. *RB*

✓B. *RAS*

C. *XPA*

D. *MSH2*

E. *HTT* (the gene associated with Huntington disease)

F. *HEXA*

G. *MECP2*

Rationale: Protooncogenes are normal cellular genes which usually play a role in cell proliferation. Mutational activation, or loss of regulation of a protooncogene causes the gene to become an oncogene. Of the choices in this question only the RAS gene is a protooncogene, so it is the best answer here. Note that in most cases, a protooncogene becomes an oncogene as a result of activation, or an increase in activity. The RB gene is a tumor suppressor gene (TSG) that regulates the cell cycle, and TP53 is a TSG that functions to suppress cell proliferation in many ways controlling cell cycle, apoptosis, and as a guardian of the genome. For a TSG to become a tumor promoter, both functional copies must be lost in the cell, or something must occur to “de-activate” the function of the gene; loss of heterozygosity is an event that can cause this deactivation. The MSH2 gene encodes a component of the mismatch repair pathway. The huntingtin gene (HTT) encodes a neuronal protein. If this gene undergoes a triplet repeat expansion (a CAG codon in exon 1), it leads to a protein with an expanded tract of glutamine amino acids, which causes the protein to attain a new function (unknown) that leads to neuronal death.

Question #: 16

An 8-year-old boy with progressive swelling of the jaw is brought to the physician by his parents. A biopsy is performed and FISH analysis of the tissue shows a translocation between chromosomes 8 and 14. Proteomic analysis of the tissue shows the elevated expression of an oncoprotein. Which of the following best describes the function of this protein?It

A. It has tyrosine kinase activity
B. It functions as a growth factor receptor
C. It activated the G1-S checkpoint
D. It is required for DNA repair
✓E. It is a transcription factor
F. It activated apoptosis

Rationale: The t(8:14) rearrangement places the MYC gene on chromosome 8 under the control of an active Ig promoter on chromosome 8, Thus there is an elevated expression of the MYC gene which is a transcription factor that regulates the expression of many genes.

Question #: 17

A 72-year-old woman comes to the physician for a follow-up appointment to discuss results of her mammogram. Her 41-year-old daughter was recently diagnosed with breast cancer. Results of the mammogram show no evidence of breast cancer. A detailed family history is obtained, and the resulting pedigree of affected family members is shown. The patient is indicated by the arrow. Genetic testing shows that the patient, her daughter and other affected family members have the same pathogenic germline mutation in BRCA2. Which of the following best describes the genetic principle illustrated in this patient?

A. Allelic heterogeneity
✓B. Incomplete penetrance
C. Heteroplasmy
D. Anticipation
E. Variable expressivity

Rationale: Mutations in the BRCA2 gene can increase the risk for breast and ovarian cancer and there are many environmental risk modifiers such as smoking, alcohol consumption and exogenous hormones.
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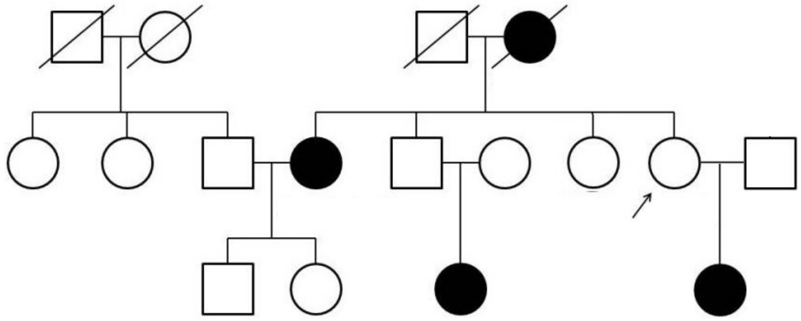

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Question #: 18

A researcher studying cancer obtains a biopsy from a 69-year-old man to study the genetic driver mutations involved in the development of the metastatic potential of the neoplasm. Next-generation sequencing shows that the cancer cells have a mutation in the gene which encodes the protein labeled "X" as shown in the figure. Which of the following best describes this protein?

A. Small monomeric G-protein
✓B. Tyrosine kinase
C. Transcription factor
D. Steroid binding protein
E. IgG Antibody
F. DNA helicase enzyme

Rationale: Receptor tyrosine kinases transduce signals from outside the cell to initiate a signal transduction cascade inside the cell. Ras is known as a small monomeric G-protein (or GTPase protein). Transcription factors are likely to activated downstream of this pathway to establish a long-term response to the signal via altered gene expression. Steroid binding proteins bind steroid hormones, so that they may become activated; then, in the nucleus, they can regulate gene expression. The IgG antibody is created by immune system cells as part of the immune response.

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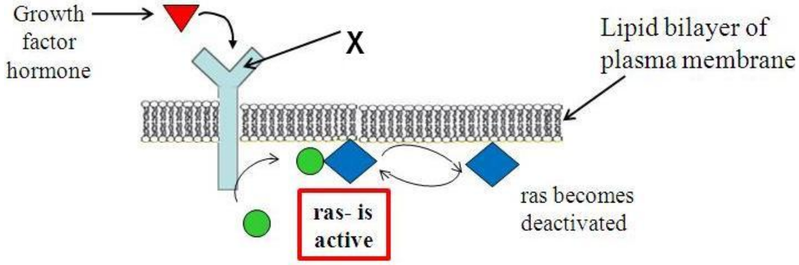

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Question #: 19

A 12-year-old boy is brought to the physician for an annual health maintenance examination. He has intellectual disability, short stature, a depressed nasal bridge, upslanting palpebral fissures, epicanthal folds, single palmar creases, and an exaggerated sandal gap on both feet. He was born with a congenital heart defect which was surgically repaired. Genetic testing when he was an infant showed that the disorder was caused by a nondisjunction event during oogenesis. Which of the following syndromes does this patient most likely have?

A. Cri-du-chat syndrome
B. Klinefelter syndrome
✓C. Down syndrome
D. Wolf-Hirschhorn syndrome
E. Angelman syndrome

Rationale: The clinical features describes Down Syndrome, and the most common way that this occurs is by nondisjunction during meiosis 1 during oogenesis and in particular for women approaching or over 40. Oocytes do not complete meiosis 1 until ovulation, and are thus "frozen" in metaphase of meiosis 1 since these cells differentiated while the female was a fetus. Therefore, the older the woman, the longer the oocyte has been kept in metaphase 1.

Question #: 20

A 70-year-old man is brought to the physician for a follow-up examiantion. He has a 6-month history of chest discomfort, trouble breathing, a worsening cough, and blood in his sputum. A CT scan show small lesions in his lungs, sputum cytology shows the presence of lung cancer cells and a diagnosis of stage 2 non-small cell lung cancer is made. The patient agrees to give a blood sample for researching biomarkers of non-small cell lung cancer before and after 2-months of chemotherapy. Tests show that the patient initially has 5-times lower concentrations of cytochrome c in his serum versus control patients. After 2-months of chemotherapy, his serum cytochrome c levels are 10-times higher than control levels. Which of the following cellular events is most likely to cause the release of this component from its normal cellular location?

A. Cleavage of procaspases
B. Formation of the apoptosome
C. Blebbing of the plasma membrane
✓D. BAX is inserted into the mitochondrial membrane
E. Increased expression of BCL2

Rationale: Cytochrome c is a mitochondrial protein that on the inner mitochondrial membrane as a peripheral protein. has been found to have dual functions in controlling both cellular energetic metabolism and apoptosis. When released out of the mitochondria through the mitochondrial permeability transition pore apoptosis may be activated via caspases cascade activation of once it is released into the cytosol. Caspases are zymogen proteins, that once cleaved become activated.

Question #: 21

An 8-year-old girl is brought to the physician by her legal guardian for a follow-up examination after a genetic test confirms a diagnosis of neurofibromatosis. A figure depicting the pathway affected by this pathogenic genetic variant is shown. Which of the following functions indicated in the figure best provides a molecular explanation of the disorder in this girl?

A. A
B. B
C. C
D. D
✓E. E

Rationale: Neurofibromin is the name of the protein product of the NF1 gene which is a tumor suppressor gene. This protein functions to inhibit the activity of RAS by promoting the intrinsic GTPase activity of RAS. In general, when RAS is bound to GTP, it is active, and cell proliferation and survival is promoted. When RAS is bound to GDP, it is not active, and proliferation, and survival are not encouraged.

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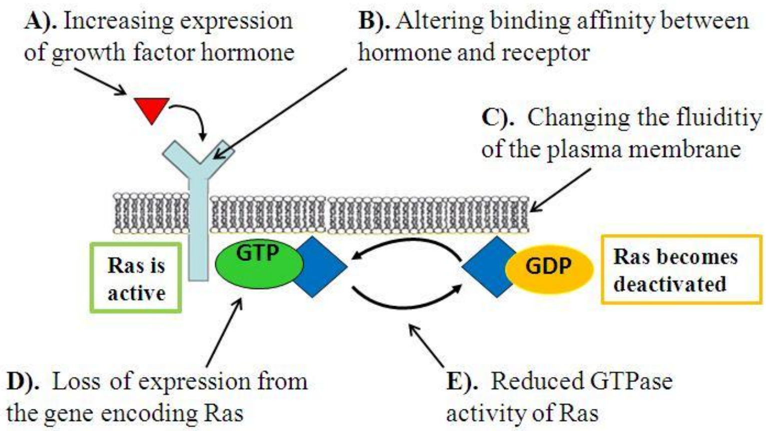

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Question #: 22

A 5-year-old girl is brought to the physician by her foster parents because of progressive impairment of muscular coordination and a multitude of surface blood vessels particularly in the eyes. Analysis of the sequence of ATM shows that the girl has compound heterozygosity for loss of function mutations in this gene. Which of the following activities is most likely to be directly compromised as a result of the genetic mutations in this patient?

A. DNA helicase activity
✓B. Detection of double strand breaks
C. Repair of pyrimidine dimers
D. Post replication mismatch repair
E. Transcription of growth promoting genes

Rationale: ATM (ataxia telangiectasia mutated) is involved in detecting DNA damage and directing the repair response. ATM is a serine/threonine protein kinase that is able to sense DNA double strand breaks, and it functions to activate other DNA repair systems by phosphorylation. Bloom syndrome is caused by loss of function of a DNA helicase. Deficiency of pyrimidine dimer repair is associated with xeroderma pigmentosa. Reduced post replication mismatch repair will cause instability of simple sequence repeats (di, tri, and tetra nucleotide repeats) which can be revealed by microsatellite instability. MYC is a classic example of transcription factor that controls the expression of many growth promoting genes.

Question #: 23

A researcher is studying genetic factors that influence the risk of multiple sclerosis. This investigator is attempting to associate the allelic variants in patients with the disorder, to compare against normal controls (individuals without the condition). Which of the following techniques was most likely used to collect the data in this analysis?

A. FISH analysis
B. Northern analysis
✓C. SNP chip
D. G-band karyotype
E. ASO blot
F. Southern blot
G. Western blot

Rationale: The vignette describes a genomic approach to a question: which areas of the genome tend to associate with patients who have a particular disorder compared to healthy controls. The only technique in the answer choices offered that can query the entire genome in a single test is the SNP chip.

Question #: 24

A researcher is attempting to catalog all genetic disorders associated with human mortality and morbidity. Which type of the heritable conditions contribute the most to the disease load in the typical human population?

A. Chromosomal aberrations
B. Autosomal dominant disorders
✓C. Multifactorial disorders
D. Digenic disorders
E. Autosomal recessive disorders
F. Mitochondrial disorders

Rationale: Multifactorial disorders exhibit a combination of distinct characteristics which are very clearly differentiated from Mendelian inheritance and do not have a single genetic cause. Multifactorial disorders can also be influenced by lifestyle and environmental factors such as exercise, diet or exposure to pollutants.

Question #: 25

A researcher studying non-syndromic congenital heart defects (CHD) finds that the population incidence of this multifactorial developmental defect is approximately 1 in 100 live births. Comparisons are made to estimate recurrence risk depending on the affected relative. Which of the following best explains an aspect of the inheritance of this disorder?

A. Recurrence risk will be higher in their male children then in their female children
✓B. Recurrence risk reduces dramatically as relatedness decreases
C. Recurrence risk will be the same as the population incidence
D. Recurrence risk will be one half of the population incidence
E. There is no risk to the children as this is a quantitative trait
F. Recurrence risk will be twice the population incidence

Rationale: A non-syndromic heart defect implies that there is not a “single-gene” or Mendelian genetic cause for the disorder. Instead, it is thought that there is likely to be a combination of many genetic factors and environmental influences to cause the disorder – multifactorial. An affected person who is more distantly related (say a cousin) contributes less risk than a first degree relative would. In other words, for multifactorial disorders, it matters more if a parent or sibling was affected, than a cousin. Some disorders such as pyloric stenosis exhibit a sex bias.

Question #: 26

A 32-year-old woman comes to the physician for a follow up consultation after being diagnosed with coronary artery disease. Research shows that compared to men, premenopausal women are typically at lower risk for this disorder because of the protective effects of estrogen. Which of the following best summarizes the recurrence risk prediction among the immediate members of this woman's family?

A. Her first degree relatives are at population risk levels
B. One of her parents must have had the condition
✓C. Her male children are at an increased risk for the disorder
D. Her female children are at a reduced risk for the disorder
E. Each of her children has a ½ risk of inheriting the disorder

Rationale: Most cases of CAD are multifactorial. Risk factors for CAD include being male and increased age. The risk for females to have CAD increases to the same risk as for males after menopause, as estrogen has a protective effect. Since the woman in this case is 32 years of age (pre-menopausal), she has more genetic factors which lowers her threshold for the disorder, therefore her children have an increased risk of having the disorder then the children of a male patient with the same disorder.

Question #: 27

An investigator is studying the cardiovascular effects of cholinergic drugs. He injects a dose of methacholine intravenously in an anesthetized dog. The dog's heart rate decreases. Which of the following best explains this finding?

A. Activation of M1 muscarinic receptors on vascular smooth muscle.
✓B. Activation of M2 muscarinic receptors on cardiac tissue
C. Activation of M3 muscarinic receptors on vascular endothelial cells.
D. Activation of nicotinic receptors on autonomic ganglia
E. Activation of presynaptic M2 muscarinic receptors in adrenergic terminals.

Rationale: M2 receptors ae coupled to Gi proteins, which interact with K+ channels. Activation of cardiac M2 receptors leads to opening of these K+ channels evoking a decrease in heart rate.

Question #: 28

A pharmacologist is studying the cardiovascular actions of a novel drug, Drug X. He administers an intravenous infusion of Drug X to an anesthetized dog. The infusion evokes an increase in both blood pressure and mean arterial pressure. Which of the following mechanisms best explains this effect?

A. Direct myocardial depression through activation of β1 receptors
B. Direct myocardial depression through activation of M2 receptors
C. Decreased heart rate through activation of β1 receptors
✓D. Constriction of blood vessels through activation of α1 receptors
E. Dilation of blood vessels in skeletal muscle through activation of β2 receptors

Rationale: Drugs that act on a1 receptors and activate them such as phenylephrine which is a selective a1 agonist or norepinephrine which is an a1, a2 and b1 agonist will evoke peripheral vasoconstriction, increase in blood pressure and mean arterial pressure. This increase will also evoke the baroreceptor reflex which causes reflex bradycardia.

Question #: 29

A 21-year-old woman comes to the physician for a routine ophthalmologic examination. A solution of phenylephrine is instilled into her eyes to dilate her pupils prior to examination. Which of the following changes is most likely to occur in the smooth muscle cells of the patient's pupil in response to the instilled drug?

A. Decreased levels of cAMP
B. Enhanced GABAergic transmission
C. Increased levels of cGMP
✓D. Increased levels of IP3
E. Decreased levels of cGMP
F. Increased levels of cAMP
G. Decreased levels of IP3

Rationale: Phenylephrine is a full agonist at alpha-1 adrenergic receptor which is coupled to Gq. Binding and activation of these receptors will increase intracellular levels of IP3 and therefore calcium that will lead to contraction of the smooth muscle cells of the eye leading to pupillary dilatation.

Question #: 30

An investigator administers an intravenous infusion of low-dose epinephrine to an anesthetized dog. The infusion evokes a decrease in diastolic blood pressure. Which of the following mechanisms best explains this effect?

A. Direct myocardial depression through activation of beta1 receptors
B. Direct myocardial depression through activation of M2 receptors
C. Decreased heart rate through activation of beta1 receptors
D. Dilation of blood vessels of skin, mucosa and kidney through activation of alpha1 receptors
✓E. Dilation of blood vessels in skeletal muscle through activation of beta2 receptors

Rationale: Low dose of epinephrine will preferentially bind to beta-2 receptors located in the smooth muscles of the vessels that supply skeletal muscle. These vessels also have alpha-1 receptors. Beta-2 receptors have higher affinity for epinephrine than alpha-1 therefore "low dose" epinephrine will activate these receptors. When these receptors are activated they will cause relaxation of the blood vessel's smooth muscle and overall vasodilatation.